Muneeb ul **Hassan**

Data Scientist · Machine Learning Engineer

Paris, France

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"Data Scientist with 7+ years of experience designing, developing, and deploying production-grade machine learning systems for the real estate domain, spanning property valuation, energy-rating estimation, and computer vision. Experienced in owning the full ML lifecycle, from problem formulation and feature engineering to model architecture, rigorous experimentation, evaluation, and deployment. Comfortable translating research ideas into reliable production systems, building custom deep-learning and gradient-boosting models, quantifying prediction uncertainty, and shipping them through a modern MLOps stack (BentoML, MLflow, Optuna, Docker). I am looking to contribute to high-impact teams building intelligent data products, where I can apply rigorous experimentation and deliver interpretable, production-ready ML solutions."

Experience

Data Scientist
HOMIWOO — *Paris, France* — *Sept. 2019 - PRESENT*

- **Built the production Automated Valuation Model (AVM)** for French residential real estate, an attention-based comparable-sales neural network (PyTorch Lightning) that estimates property price as a learned, attention-weighted aggregate of comparable transactions and listings, with bounded per-comparable price adjustment for built-in interpretability.
- **Designed the uncertainty-estimation layer** using Conformalized Quantile Regression to produce calibrated prediction intervals and a confidence score derived from the model's own internal attention and adjustment signals.
- **Developed the DPE/GES energy-rating estimation service**, gradient-boosted models (XGBoost) predicting energy consumption and greenhouse-gas emissions, with surface-aware conversion to the regulatory A–G class and a custom asymmetric loss to control over/under-prediction bias.
- **Engineered spatial neighbour-based feature pipelines**, using similarity-weighted aggregations (distance, surface, construction year, recency) of nearby properties' known DPE ratings as predictive features for energy-class estimation.
- **Computer-vision research** for real-estate listing-photo analysis, including object detection (YOLOv5), panoptic segmentation (OneFormer), monocular depth (DPT), image-pair matching (LoFTR) and CLIP-based classification toward floor-plan reconstruction from photos.
- **Built and operated the ML infrastructure**, packaging models as BentoML services, tracking experiments and model versions in MLflow, and running notebook-driven experimentation with A/B candidate-vs-reference evaluation and training/serving-skew validation.
- **Owned the full ML lifecycle**, dataset creation, hyperparameter optimization (Optuna), evaluation pipelines, model explainability (SHAP / Captum), and deployment of production models.

Deep Learning Research Intern
LIG Lab — *Grenoble, France* — *Feb. 2019 - JUNE 2019*

- Worked on Explainable Deep Learning for Multimedia Indexing and Retrieval and conducted a survey on Explainable AI.
- Proposed a technique; paper accepted at the CBMI 2019 conference.

Software Engineering Intern
Fujitsu — *Augsburg, Germany* — *Sept. 2017 - Sept 2018*

- Automated and optimized (TAO) QA server tests for low/mid/high-end Fujitsu servers, implementing new functionality and bug fixes in Java/Python in an Agile (Scrum) environment.

Machine Learning Research Intern
Melbourne, Australia

RMIT UNIVERSITY
Jan. 2017 - June 2017

- Optimised deep-learning hyper-parameters using hyper-heuristics for classification.
- Published the work as a paper at the A-rank ICCS-2018 conference.

Software Engineer
Karachi, Pakistan

CDC HOUSE
Jan. 2015 - May. 2015

- Developed new functionalities and fixed bugs in the production system.
- Wrote data queries to meet day-to-day business requirements.

Technical Skills

Languages	Python (Expert), SQL, Java, C/C++
Machine Learning	Deep Learning, Attention Models, Gradient Boosting (XGBoost, LightGBM, CatBoost), Ensembles, Feature Engineering, Hyperparameter Optimization
Computer Vision	Object Detection (YOLO), Semantic / Panoptic Segmentation (OneFormer), Monocular Depth (DPT), Feature Matching (LoFTR), CLIP
Frameworks	PyTorch, PyTorch Lightning, Scikit-learn, XGBoost, TensorFlow
Data	NumPy, Pandas, Elasticsearch
MLOps	BentoML, MLflow, Optuna, Docker, DVC, Dagster, Git
Modeling	Uncertainty / Conformal Prediction, Explainable AI (SHAP, Captum), A/B Evaluation, Time-Series

Education

Master of Informatics (Specialization in Data Science)
Grenoble, France

ENSIMAG, GRENOBLE INP
Sept 2018 - Present

Master of Computer Science
Melbourne, Australia

RMIT UNIVERSITY
Jan 2016 - July 2017

B.E in Computer and Information Systems Engineering
Karachi, Pakistan

NEDUET
2011 - 2014

Publications

ul Hassan, Muneeb, Nasser R. Sabar, and Andy Song. “Optimising Deep Learning by Hyper-heuristic Approach for Classifying Good Quality Images.” *In International Conference on Computational Science*, pp. 528-539. Springer, Cham, 2018.

Muneeb ul Hassan, Philippe Mulhem, Denis Pellerin, and Georges Quénot. “Explaining visual classification using attributes.” In 2019 International Conference on Content-Based Multimedia Indexing (CBMI). IEEE, 2019.

Languages

English	Fluent
French	B1 - B2 (actively improving)
Urdu/Hindi	Native